

Lab 8 for Batch G3

Dated: 28-09-2026

1. Write a C/C++ program to construct and compare LR(0) and SLR(1) parsing tables for the grammar:

$S \rightarrow id = E$

$S \rightarrow E$

$E \rightarrow * E$

$E \rightarrow id$

Your program must:

1. Number and augment the grammar.
2. Generate the complete set of **LR(0) items**.
3. Construct the LR(0) ACTION/GOTO table.
4. Detect and report all conflicts.
5. Compute the FOLLOW sets required for SLR(1).
6. Construct the SLR(1) ACTION/GOTO table.
7. Identify which LR(0) conflicts are resolved by SLR(1).
8. Implement both parsers.

Instructions

Due date: 05.10.2026, 11:30 P.M for Batch G1 (30 minutes extra is given for the documentation)

Cut-off date: 06.10.2026, 11:50 P.M. (Late submission deadline)

Note:

1. If the submission is after the due date/time, marks will be deducted for every 6 hours till the cut-off date. After the Cut-off date, No evolution and no marks
2. If the output is not shown for any of the above components, the mark cannot be given even if it is included in your documentation.